

Quantum Gravitation

Lecture 14

Final Project: Joint Inference

- Degrees of freedom
- Quantization
- Spectrum and response
- Consistency checks
- Example

■ Degrees of freedom

We separate propagating polarizations from constrained components before quantization.

$$L = \chi_{scat}^2 + \chi_{prop}^2 + \chi_{therm}^2$$

■ Quantization

Canonical normalization is fixed before creation and annihilation operators are introduced.

$$accept = algebra \wedge registry \wedge resource$$

■ Spectrum and response

Poles, residues, and group velocity are read from the same quadratic operator.

$$r_* = \operatorname{argmin}_r(L_r, d_r, \varepsilon_{\text{num},r})$$

■ Consistency checks

Hermiticity, positive norm, source conservation, and the low-energy limit are checked explicitly.

■ Example

A short numerical estimate fixes the scale of the effect before a full fit is attempted.

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Reading: Final Project Protocol QG-30; Q-Medium authoring standard; RENORM-66 batch guide